TR-SPAD digit classification

Gamma–exponential BOCPD and adaptive integration

Problem formulation

One simulated TR-SPAD recording contains a variable number of photon detections.

INPUT

$$S = \{(x_i, y_i, t_i)\}_{i=1}^{N}$$

Pixel coordinates: 0 to 27
Arrival times: seconds
Acquisition interval: $[0, T_{acq}]$

OBJECTIVE

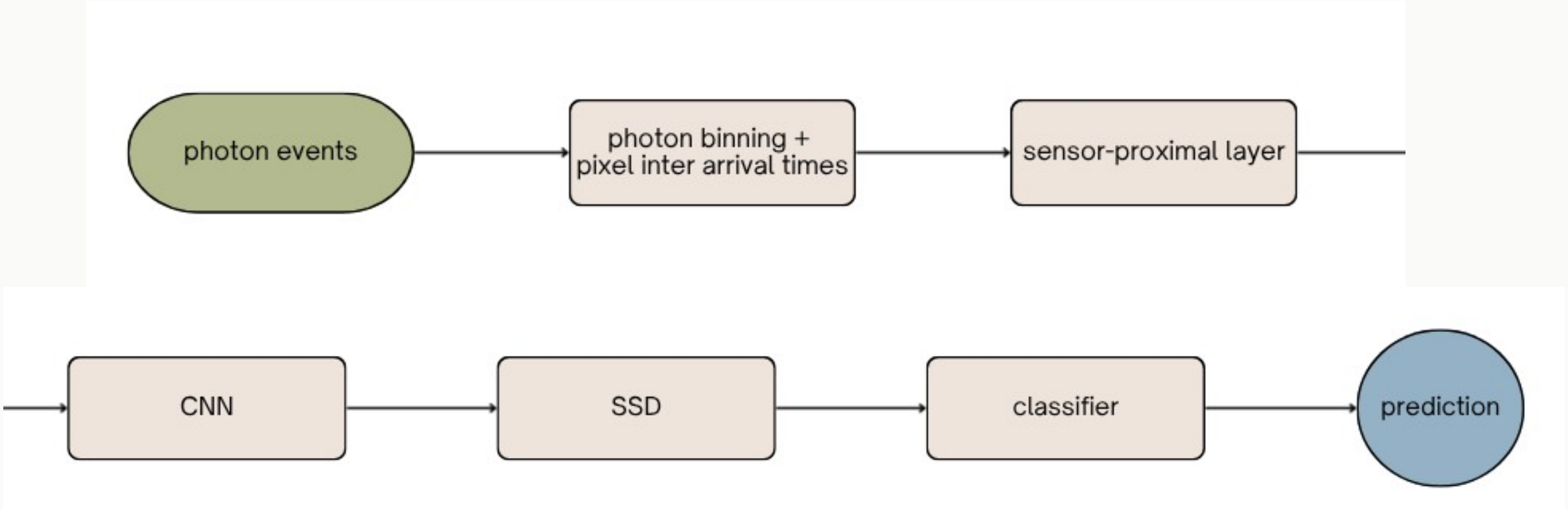
Classify the digit from the photon stream

Use photon timing in a CNN-based pipeline that generalizes to unseen recordings.

The sensor-proximal layer supplies spatial readouts without a separate conventional reconstruction stage.

Pipeline overview

The sensor-proximal layer processes all events. The neural network receives selected spatial readouts.



Photon-event tensor

Each recording uses a dense event-slot representation.

Δt cube [H, W, T, K]

Dimension	Description	Experiment
H, W	Pixel location	28 × 28 pixels
T	Equal temporal bins	$\Delta t_{bin} = T_{acq} / T$
K	Photon slots per pixel and bin	Maximum occupancy in the recording

A binary mask has the same shape. With batch size one, the implementation uses [1, H, W, T, K].

Binning preserves interarrival times

Illustrative single pixel, with 10 µs bins and acquisition starting at t = 0.

Arrival times: 4, 9, 13, 29 µs

Temporal bin	Photon times	Stored Δt slots	Mask
[0, 10) µs	4, 9	[4, 5] µs	[1, 1]
[10, 20) µs	13	[4, 0] µs	[1, 0]
[20, 30) µs	29	[16, 0] µs	[1, 0]

E.g. the interval at 29 µs is 29 − 13 = 1

Photon arrival model

Assume a locally constant detection rate at each pixel.

OBSERVED INTERVAL

$$\Delta t_i = t_i - t_{i-1}$$

$$\Delta t_i \mid \lambda \sim \text{Exponential}(\lambda)$$

$$f(d \mid \lambda) = \lambda \exp(-\lambda d)$$

UNCERTAINTY ABOUT THE RATE

$$\lambda \mid \alpha, \beta \sim \text{Gamma}(\alpha, \beta)$$

α: shape
β: rate parameter, in seconds
Mean photon rate: α / β

What each forecaster stores

Up to eight hypotheses represent different possible times of the latest rate change.

Stored quantity	Description	Used for
Birth bin / age r_s	How long this rate regime may have lasted	Integration-time estimate
Gamma parameters α_s, β_s	Uncertainty about the photon rate	Predictive likelihood
Normalized weight ν_s	Relative support for this hypothesis	Weighted run-length estimate

Rate uncertainty and hypothesis weight play different roles.

Update at a new photon

Use the full interval d since the previous photon at this pixel.

SCORE THE INTERVAL

$$L_s(d) = \frac{\alpha_s \, \beta_s^{\alpha_s}}{(\beta_s + d)^{\alpha_s + 1}}$$

UPDATE HYPOTHESIS SUPPORT

Unnormalized continuing weight = $(1 - \gamma)\, \nu_s\, L_s(d)$

Include the new-change candidate, then normalize retained weights.

UPDATE THE CONTINUING RATE POSTERIOR

$$\alpha_s \leftarrow \alpha_s + 1 \qquad \beta_s \leftarrow \beta_s + d$$

Run length becomes an integration time

Compute one adaptive time constant per pixel at the beginning of each raw bin.

WEIGHTED AGE ESTIMATE

$$\hat{r} = \exp[\ \Sigma_s\ \nu_s\ \ln(1 + r_s)\ ] - 1$$

$\hat{r}$ is in “bin units”, to be usable for integration we first minpool over the local neighborhood and then convert it into seconds.

$$\tau_n = \text{spatially pooled } \hat{r}_n \times \Delta t_{bin}$$

A positive lower bound on τ avoids division by zero.

Adaptive exponential integration

A photon adds a contribution. Elapsed time decays the existing accumulator.

$$I \leftarrow \exp(-\delta t / \tau_n)\, I + 1 / \tau_n$$

Short τ

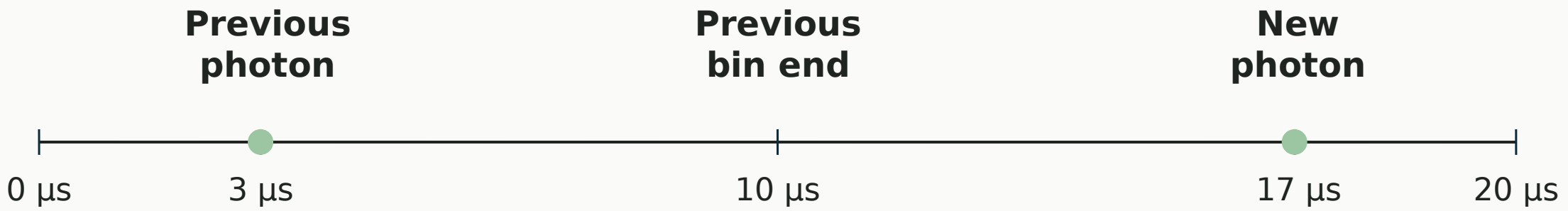
Faster response and less temporal averaging

Long τ

Longer memory and stronger temporal averaging

Two clocks at each pixel

BOCPD and the accumulator need different elapsed-time measurements.



Clock	Interval at the new photon	Purpose
Last photon	$17 - 3 = 14$ µs	BOCPD likelihood and update
Last accumulator update	$17 - 10 = 7$ µs	Exponential decay

Temporal subsampling

D sets the readout cadence. All raw bins still update the sensor-proximal state.

Illustrative configuration	Value
Acquisition duration T_{acq}	1 ms
Raw bins T and bin duration	100 bins, 10 µs each
Subsampling factor D	20 bins per readout
Readout interval and count	200 µs, 5 readouts

BOCPD + integrator

Process every event and every bin end

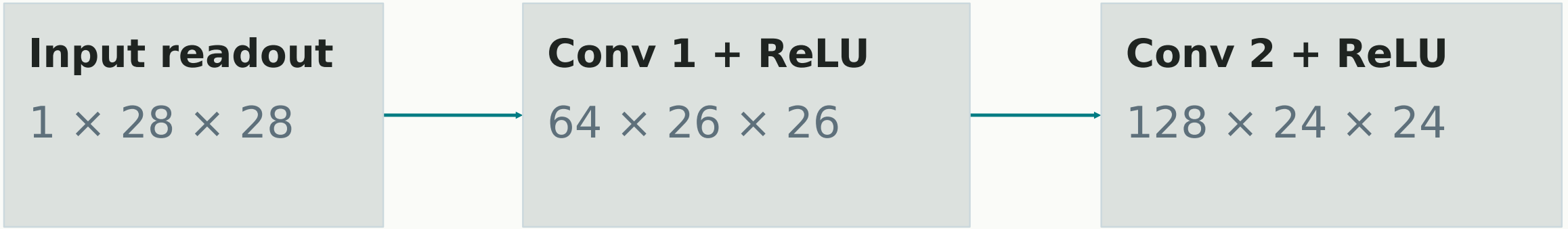
CNN + SSD

Process saved readouts

Readout times in this example: 0.2, 0.4, 0.6, 0.8 and 1.0 ms.

Spatial feature extraction

Each normalized integration readout enters the CNN as a single-channel image.

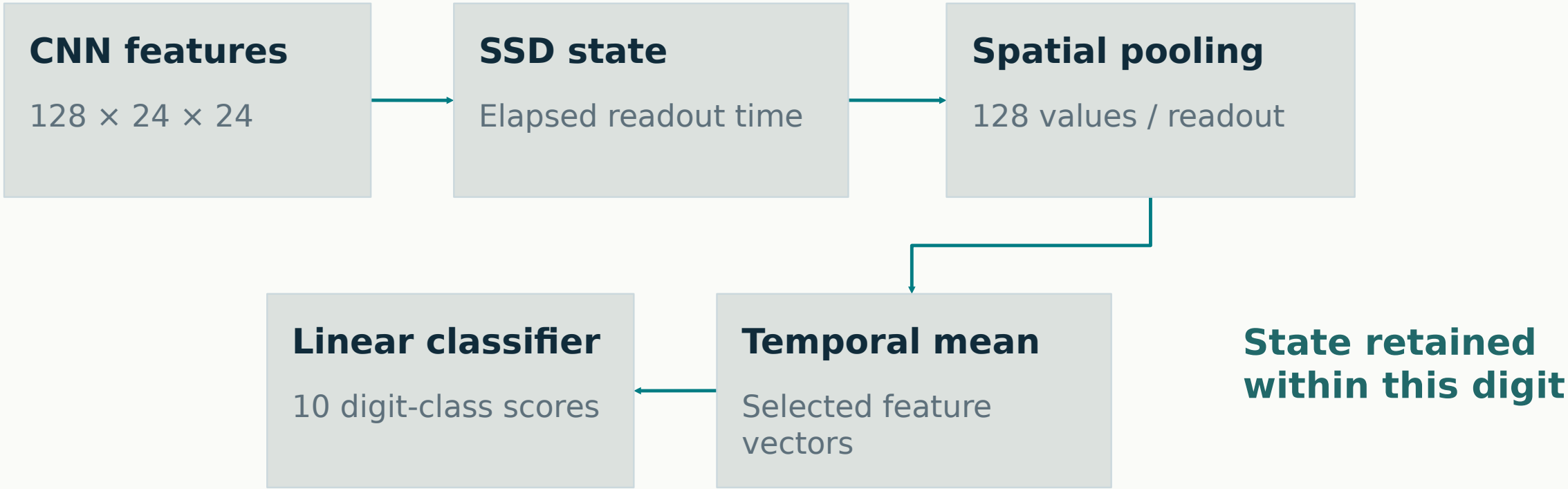


BOTH CONVOLUTIONS

3 × 3 kernel
Stride 1
No padding

Temporal features and classification

The SSD layer combines each readout with state from preceding readouts.



Training and evaluation

Both modes reset the model state between independent digit recordings.

Operation	Training	Evaluation
Input delivery	Whole recording	Successive packets of new events
Within the digit	Process all bins chronologically	Retain state between packets
Prediction features	Selected readouts of the clip	Available readouts in the history window
At the next digit	Fresh state	Fresh state

A clip ends at the acquisition boundary, even if its final bins contain no photons.

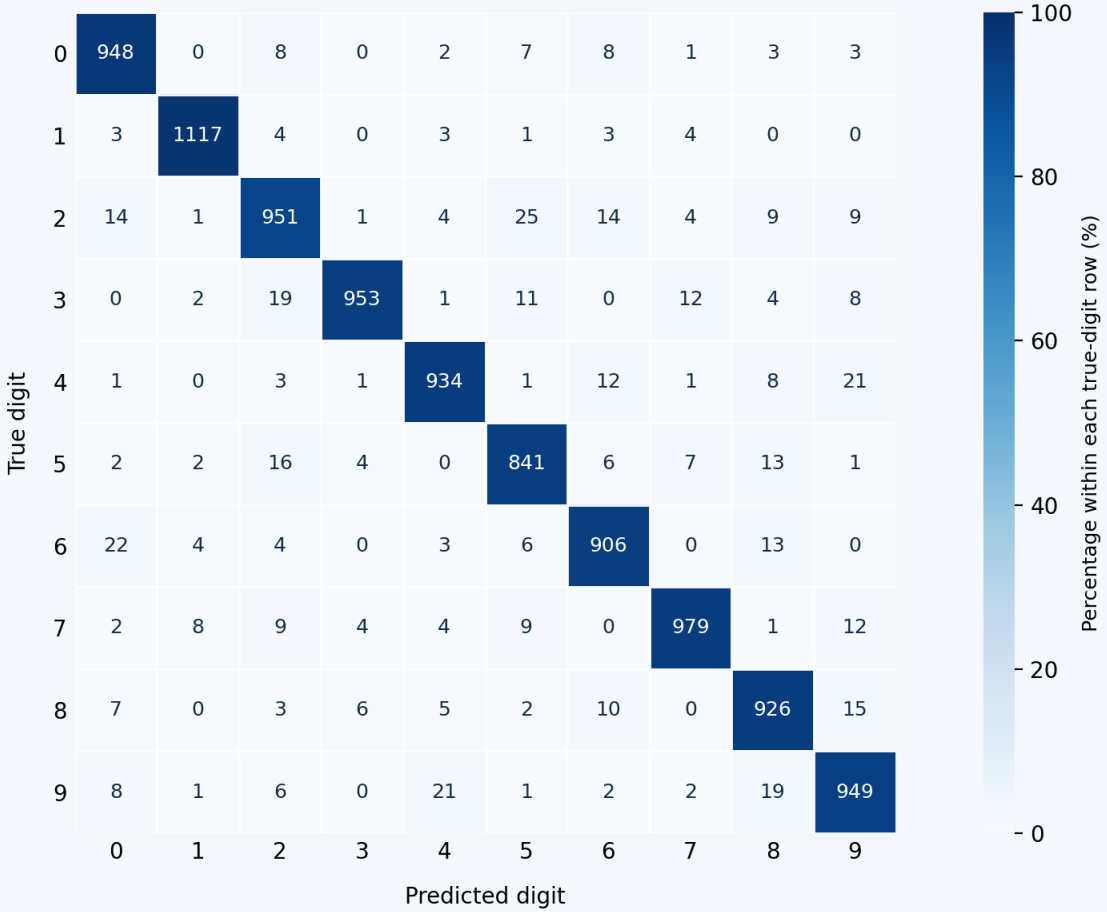
TR-MNIST | Camera evaluation

Final recording predictions and accuracy across all camera readouts

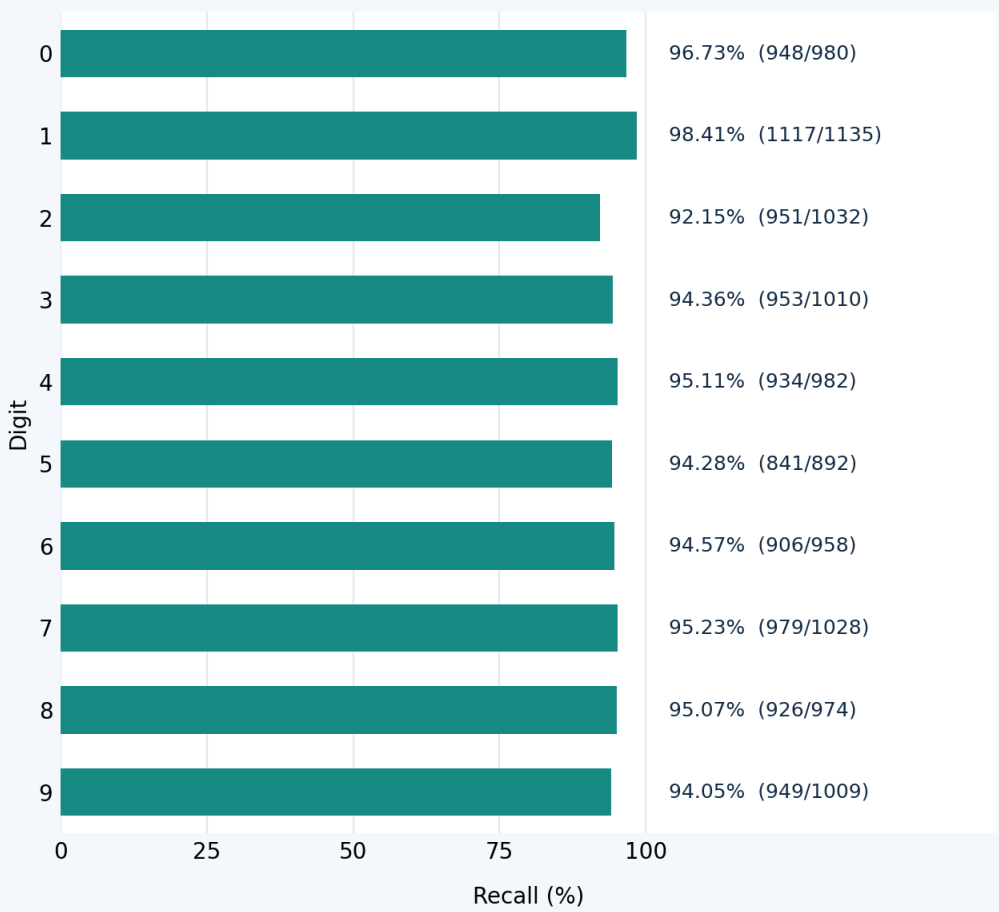
2650mlux evaluation

SAMPLES TESTED	CORRECT FINAL PREDICTIONS	FINAL ACCURACY	ALL-READOUT ACCURACY	FINAL MEAN CROSS-ENTROPY
10,000	9,504	95.04%	93.68%	0.1670

Confusion matrix | Final readout



Per-digit recall | Final readout



Matrix labels are counts; color shows row percentages. Final metrics use one prediction per recording.

Camera readouts: 50,000 | Correct readouts: 46,840. All-readout accuracy includes intermediate and final predictions.

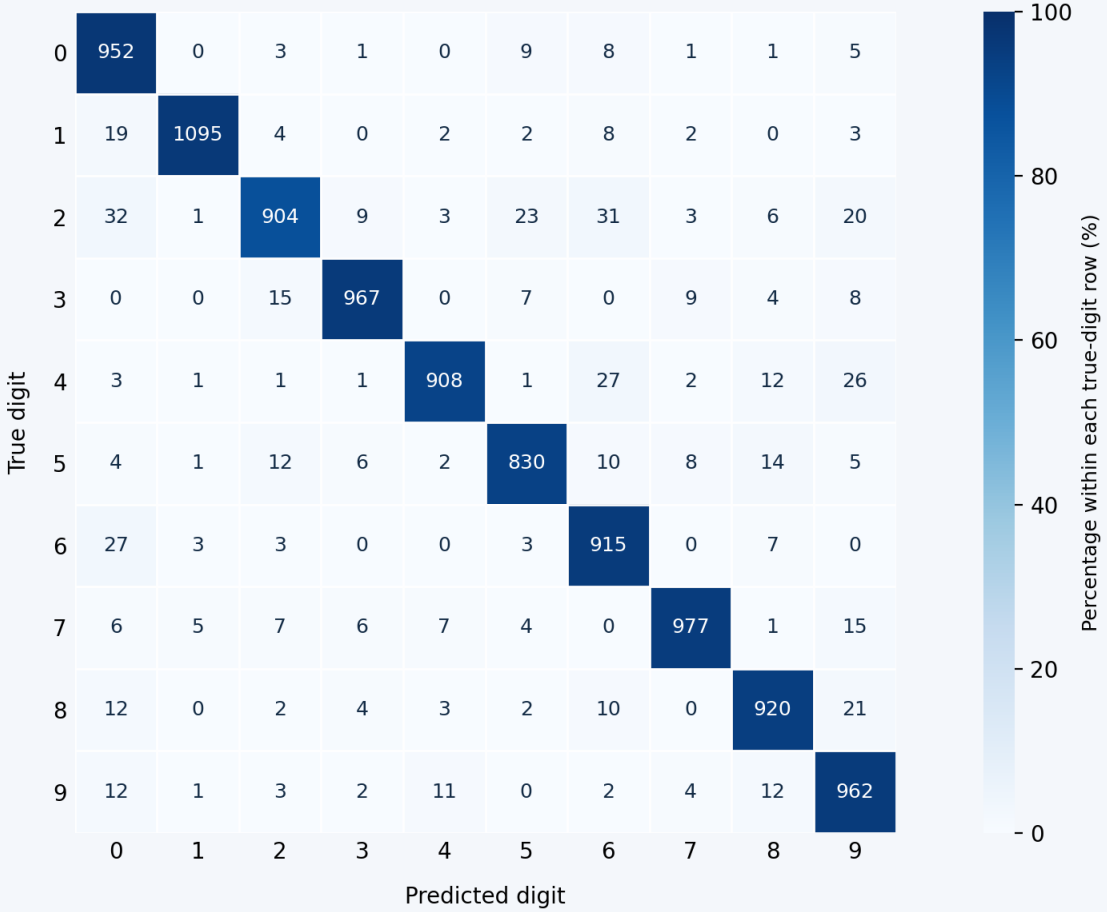
TR-MNIST | Camera evaluation

Final recording predictions and accuracy across all camera readouts

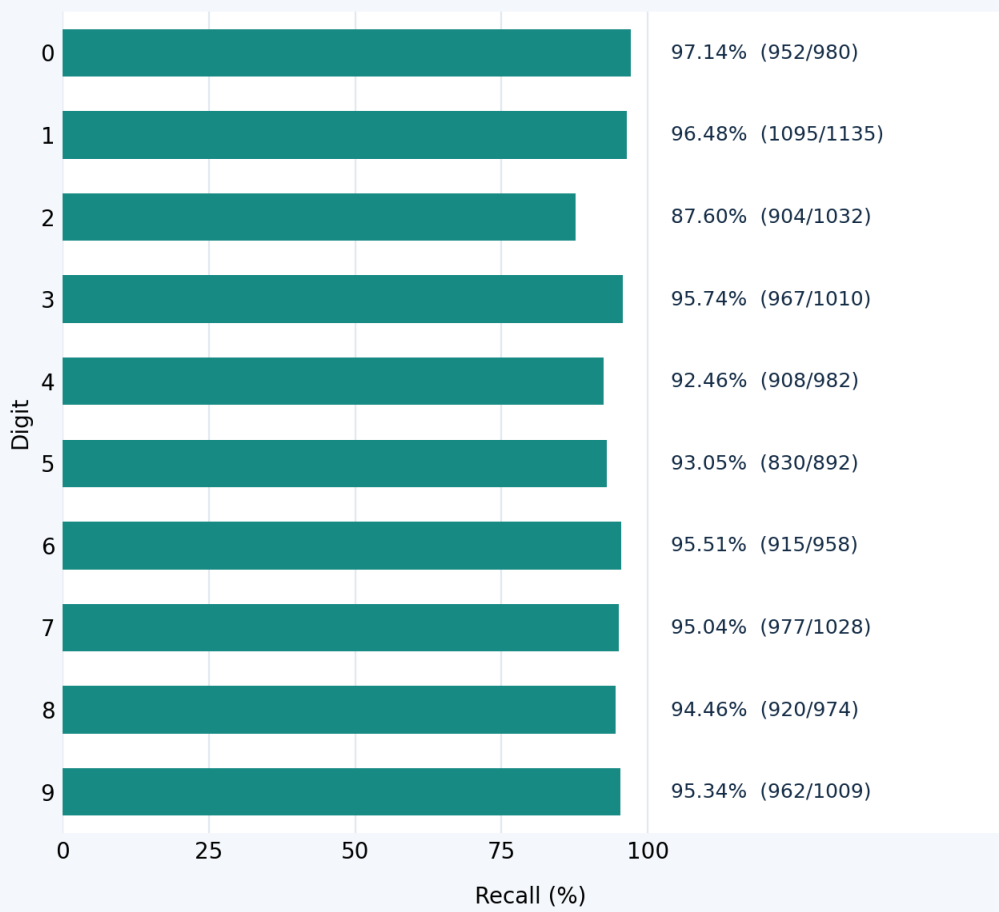
640mlux evaluation

SAMPLES TESTED	CORRECT FINAL PREDICTIONS	FINAL ACCURACY	ALL-READOUT ACCURACY	FINAL MEAN CROSS-ENTROPY
10,000	**9,430**	**94.30%**	**94.42%**	**0.1805**

Confusion matrix | Final readout



Per-digit recall | Final readout



Matrix labels are counts; color shows row percentages. Final metrics use one prediction per recording.

Camera readouts: 50,000 | Correct readouts: 47,208. All-readout accuracy includes intermediate and final predictions.

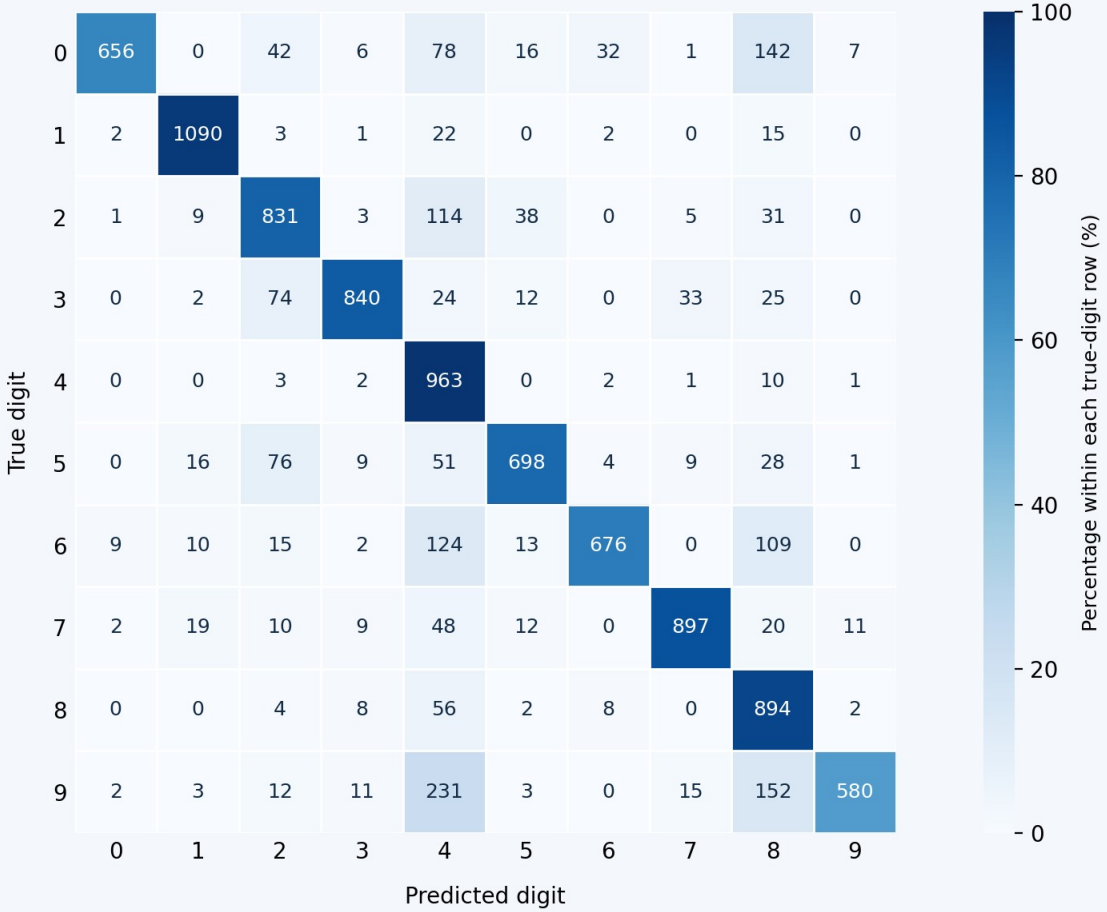
TR-MNIST | Camera evaluation

Final recording predictions and accuracy across all camera readouts

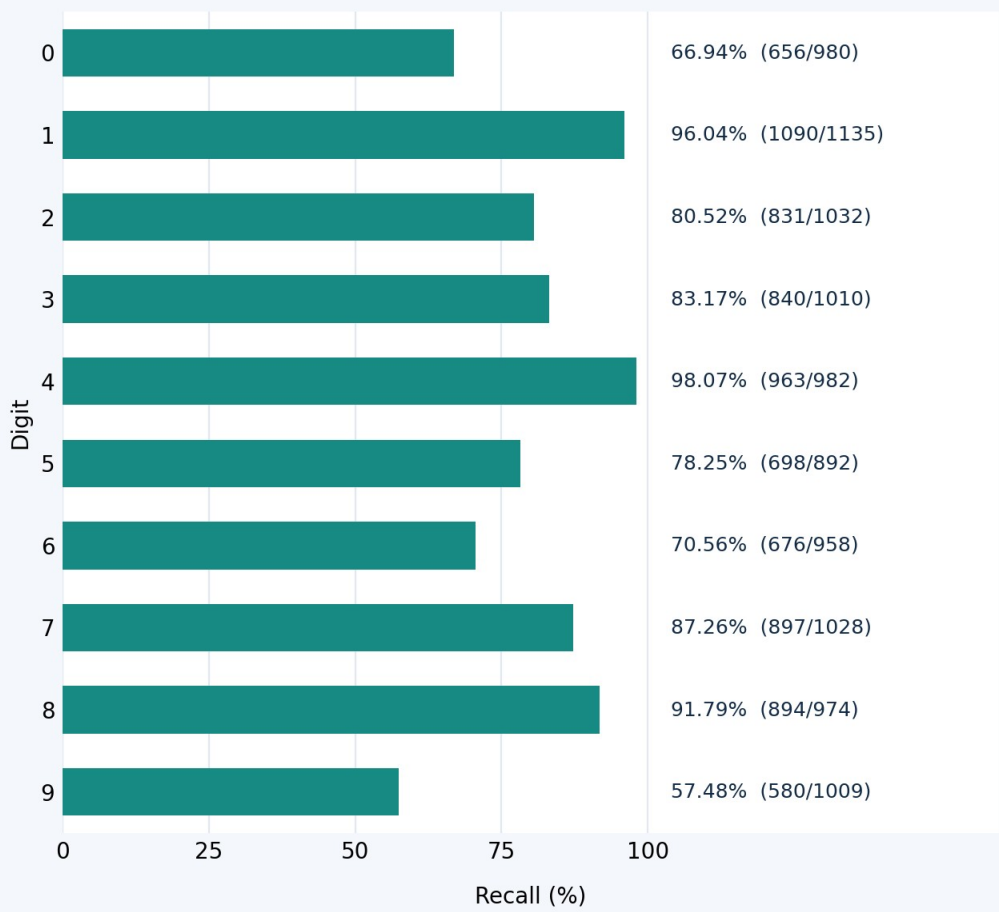
160mlux evaluation

SAMPLES TESTED	CORRECT FINAL PREDICTIONS	FINAL ACCURACY	ALL-READOUT ACCURACY	FINAL MEAN CROSS-ENTROPY
10,000	**8,125**	**81.25%**	**63.36%**	**0.5932**

Confusion matrix | Final readout



Per-digit recall | Final readout



Matrix labels are counts; color shows row percentages. Final metrics use one prediction per recording.

Camera readouts: 50,000 | Correct readouts: 31,682. All-readout accuracy includes intermediate and final predictions.